

Grouping the elements

Chemistry is the science that seeks to understand how elements behave and how they combine into compounds. In order to achieve this, chemists began trying to organize all known elements according to their physical and chemical properties. Over 200 years later, the result is the modern periodic table, which arranges elements by the similarity of their properties.

Antoine Lavoisier

In 1789, the French scientist Antoine Lavoisier produced an early list of chemical elements, which he called “simple substances”. Lavoisier’s list had 55 entries, divided into metals, non-metals, and gaseous elements, such as oxygen.



Drops of the liquid metal mercury



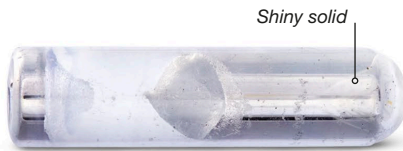
Pure oxygen gas in a glass sphere



Yellow crystals of the non-metal sulfur in rock

Johann Döbereiner

The German chemist Johann Döbereiner showed in 1817 that most elements could be grouped into threes (or “triads”). Some elements were “missing” at the time as they had not yet been discovered, and this finding encouraged the search for new elements in the 19th century. Elements in each triad – such as lithium, sodium, and potassium – shared chemical properties.



Potassium kept inside an airless glass vial